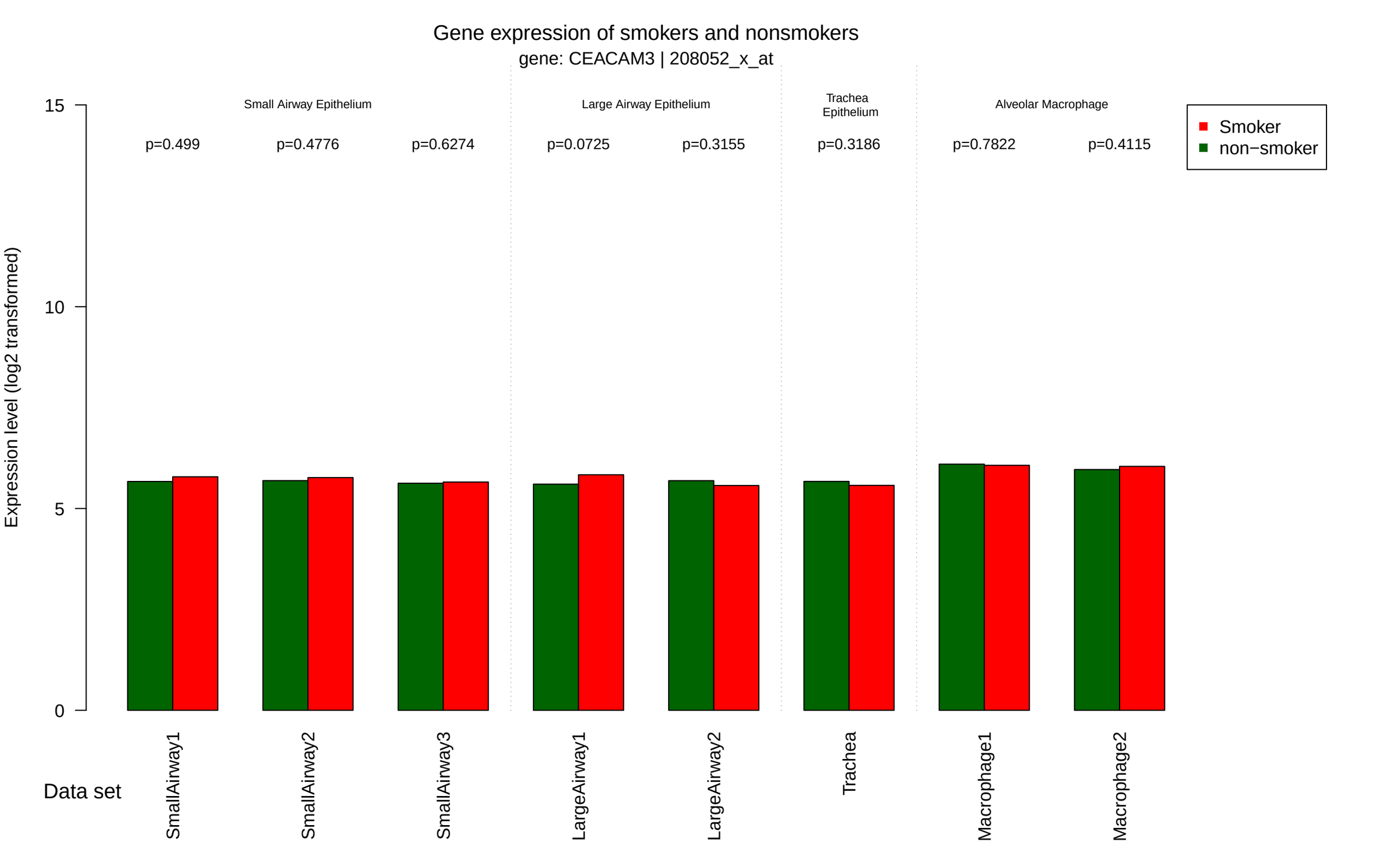
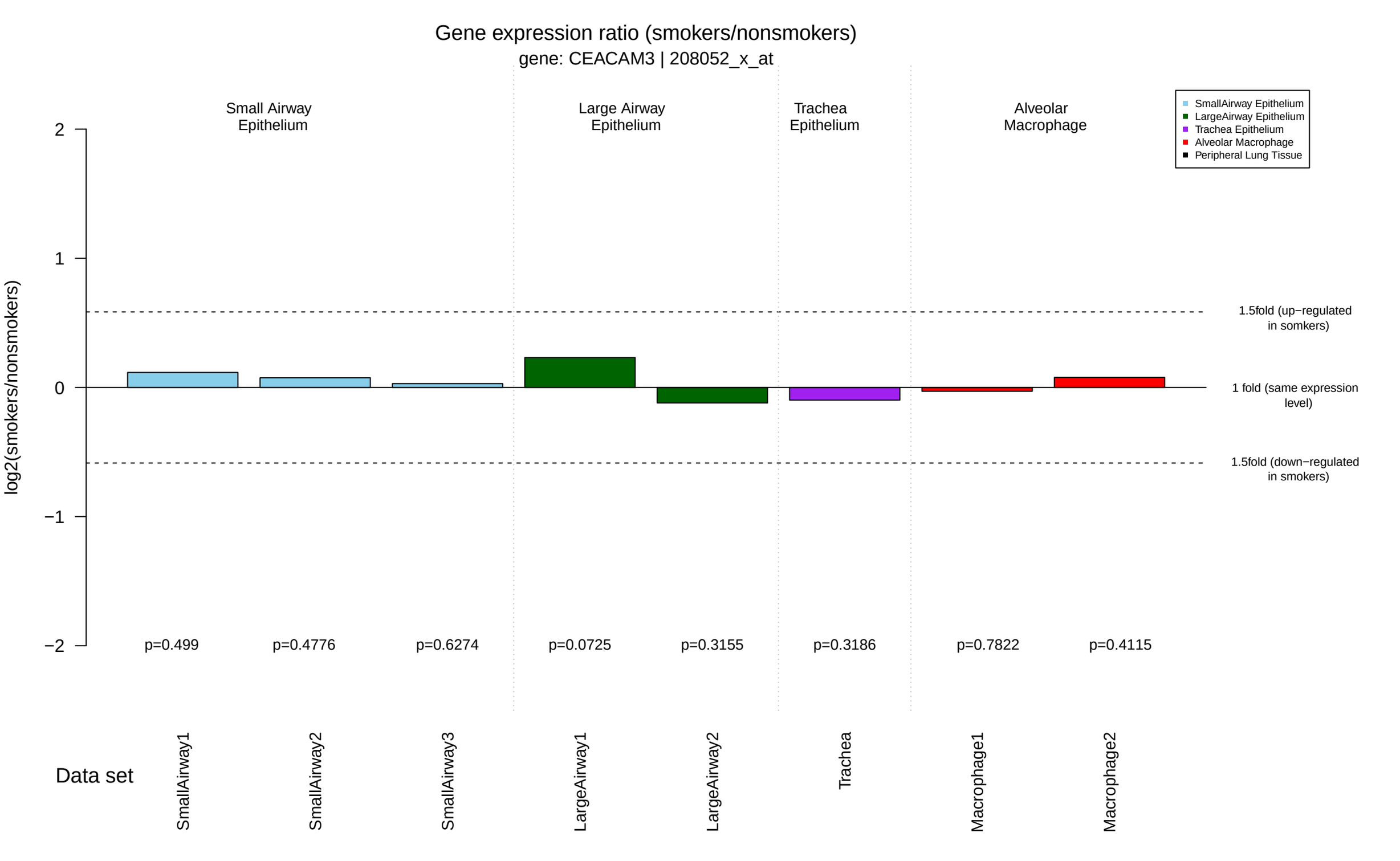
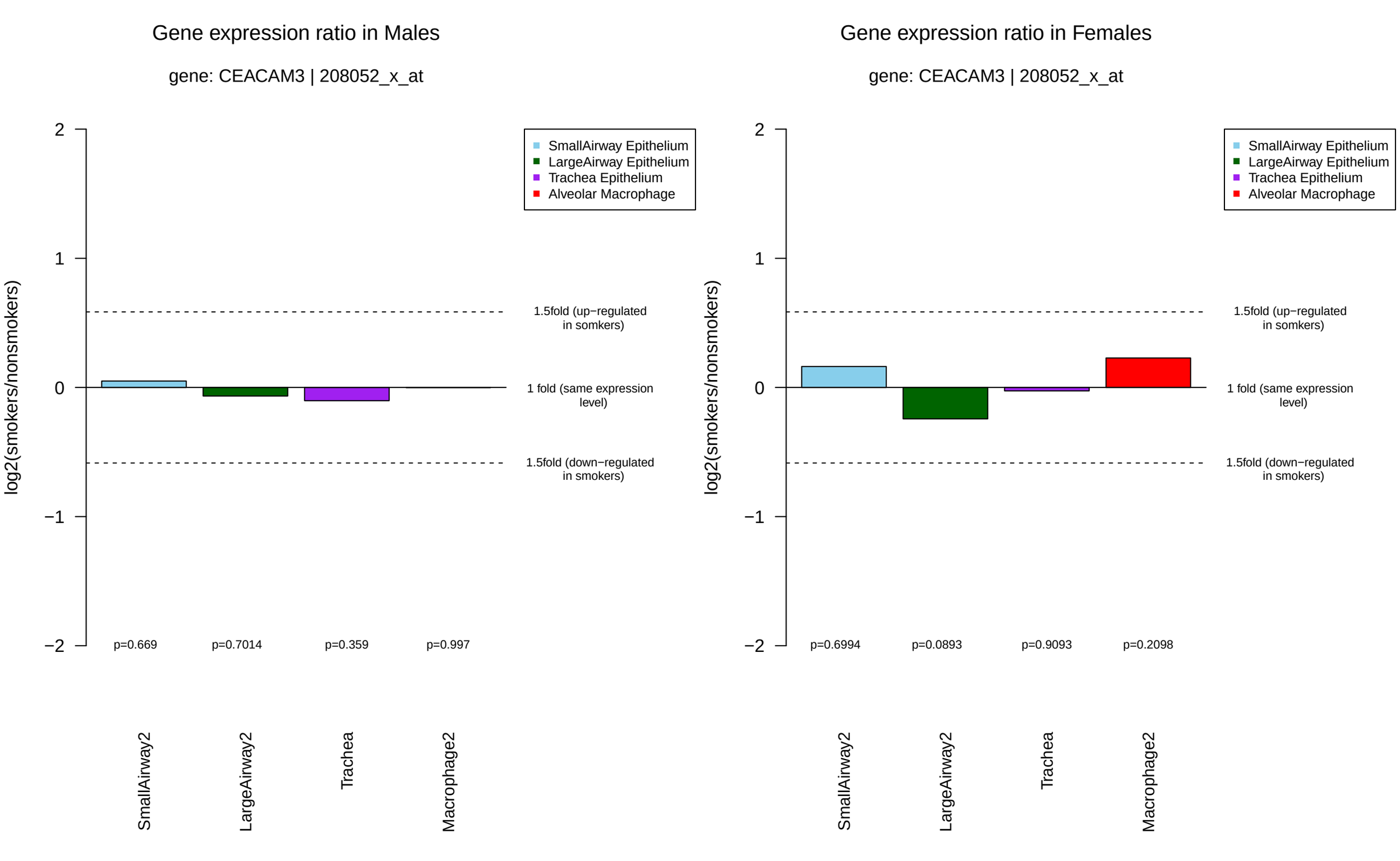
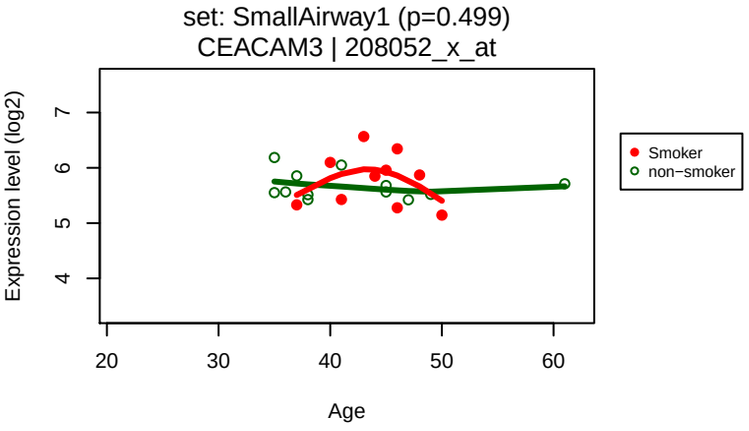
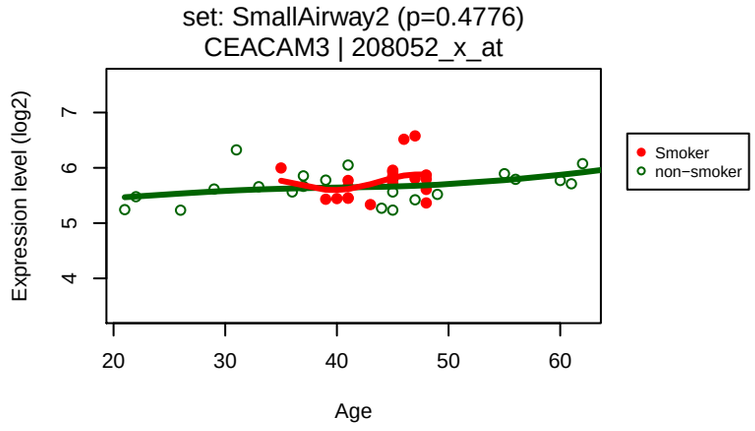
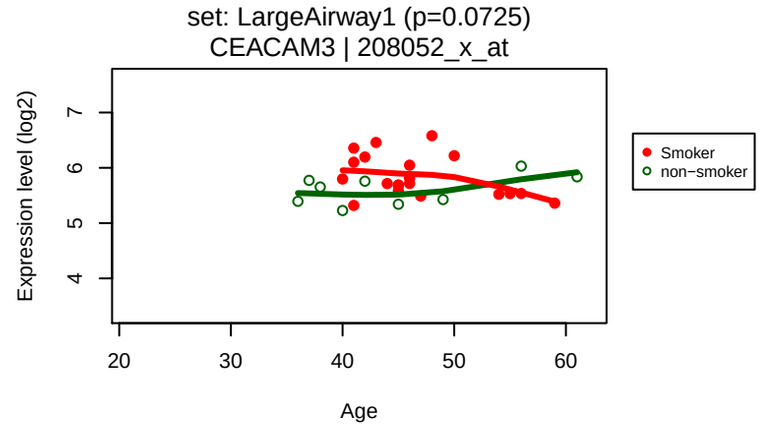

Gene expression of smokers and nonsmokers
gene: CEACAM3 | 208052_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.499
p=0.4776
p=0.6274
p=0.0725
p=0.3155
p=0.3186
p=0.7822
p=0.4115
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

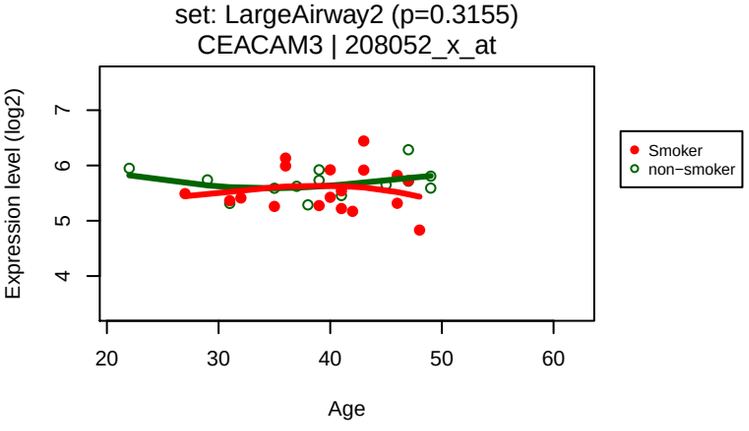

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM3 | 208052_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.499
p=0.4776
p=0.6274
p=0.0725
p=0.3155
p=0.3186
p=0.7822
p=0.4115
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

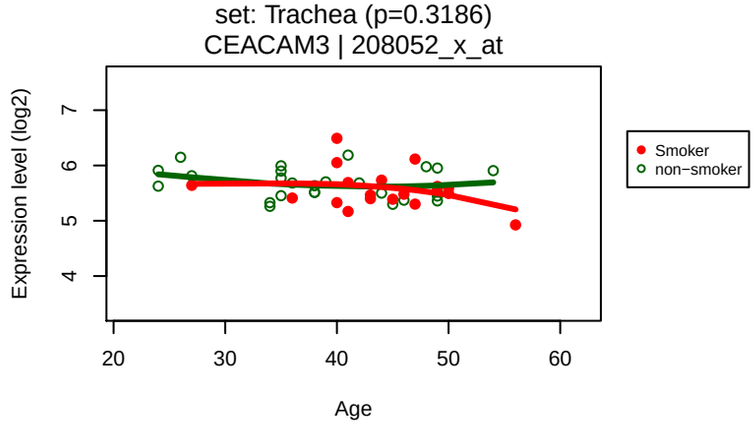

Gene expression ratio in Males
gene: CEACAM3 | 208052_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.669
p=0.7014
p=0.359
p=0.997
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM3 | 208052_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6994
p=0.0893
p=0.9093
p=0.2098
SmallAirway2
LargeAirway2
Trachea
Macrophage2

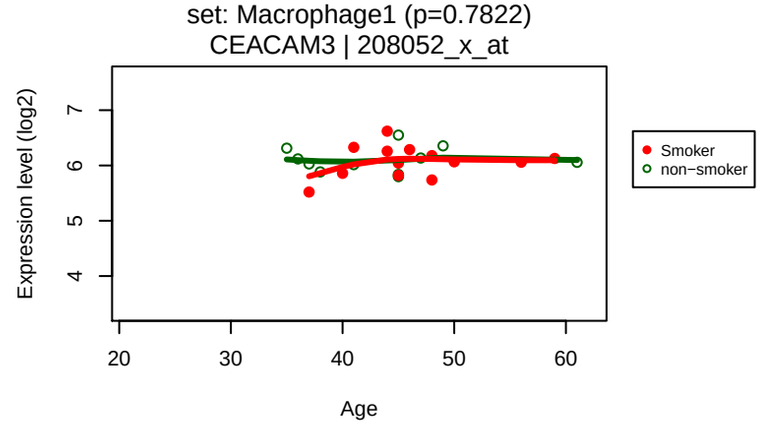

set: SmallAirway1 (p=0.499)
CEACAM3 | 208052_x_at
Expression level (log2)
Age
Smoker
non−smoker

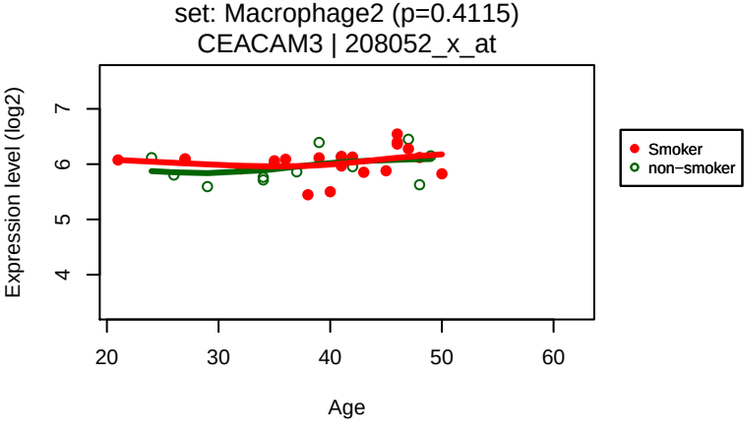

set: SmallAirway2 (p=0.4776)
CEACAM3 | 208052_x_at
Expression level (log2)
Age
Smoker
non−smoker

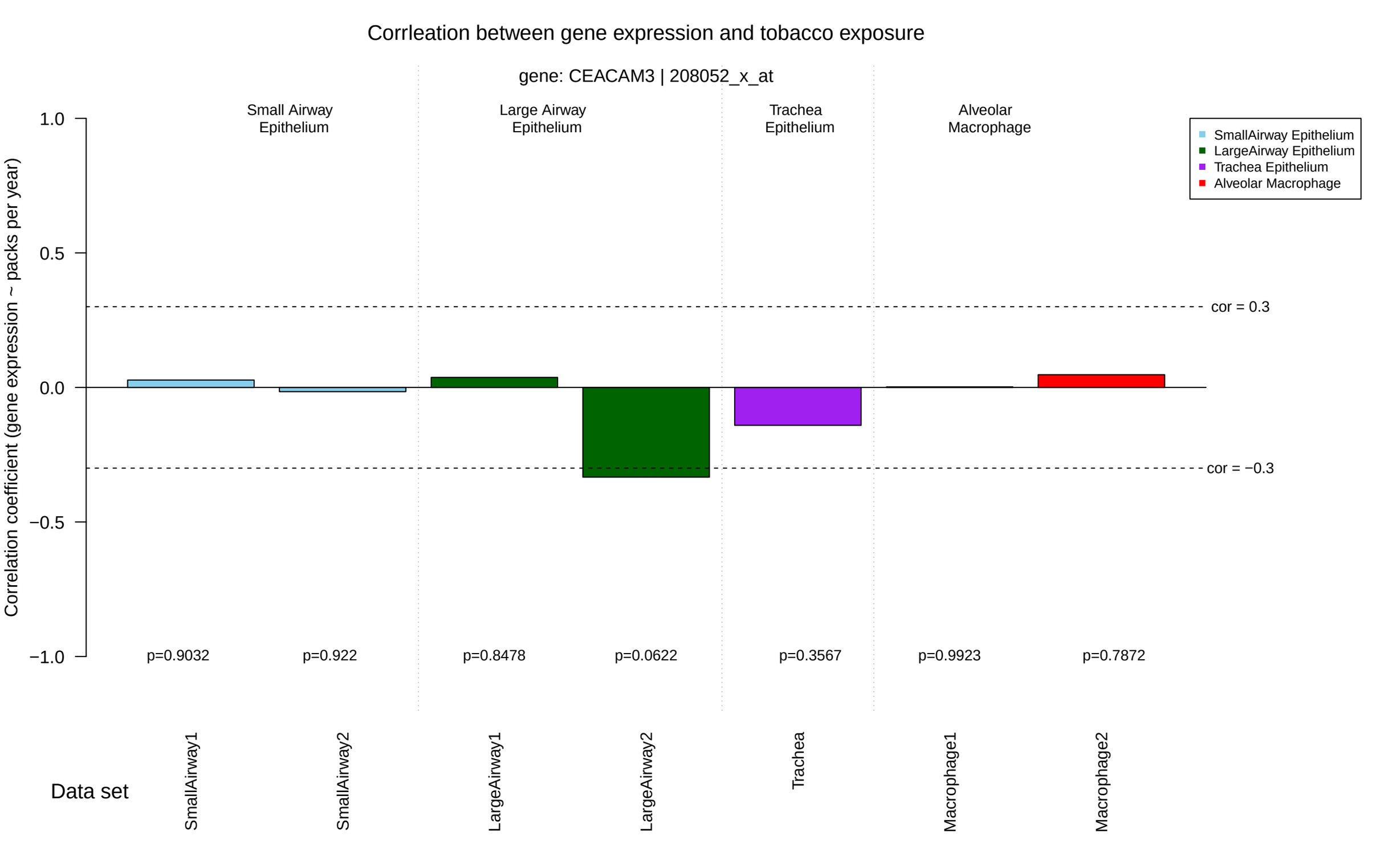

set: LargeAirway1 (p=0.0725)
CEACAM3 | 208052_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3155)
CEACAM3 | 208052_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3186)
CEACAM3 | 208052_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7822)
CEACAM3 | 208052_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4115)
CEACAM3 | 208052_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM3 | 208052_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9032
p=0.922
p=0.8478
p=0.0622
p=0.3567
p=0.9923
p=0.7872
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2